

## Education

University of Utah  
B.S. Computer Science

Salt Lake City, UT  
August 2020 – May 2025

**Relevant Coursework:** Computer Systems, Machine Learning, Computer Graphics, Algorithms, Software Practice I & II, Database Systems, Computer Networks, Foundations of Data Analysis, Models of Computation, and Linear Algebra.

## Skills

**Languages:** Python, TypeScript, C++, SQL, Java

**Software:** Pandas, NumPy, React, Node.js, AWS, Azure, Docker, Next.js, .NET, REST, Django, MongoDB, Linux

## Experience

L3Harris Technologies  
Associate Software Engineer

Dallas, TX  
June 2025 – Present

- Developed signal processing modules in Python and C++ on embedded systems, collaborating with hardware teams to optimize performance and resource utilization.
- Built microservices for high-frequency data ingestion, processing, and storage, leveraging Podman and serverless functions.
- Automated CI/CD testing with pipelines, integrating static code analysis and integration testing.

Doxy.me  
Software Engineer

Charleston, SC  
August 2024 – May 2025

- Owned product features end-to-end across web and mobile surfaces, building React and TypeScript frontend components with Node.js backend APIs that powered real-time clinical video sessions via WebRTC for thousands of daily users, from design through production.
- Improved user experience quality by iterating directly with clinical teams to translate real usage feedback into polished, intuitive interfaces, shortening the loop between user signal and shipped improvements.
- Accelerated release velocity by maintaining CI/CD pipelines and AWS infrastructure automation, enabling the team to ship high-quality product updates multiple times per week without stability tradeoffs.

HEXstream  
Software Engineering Intern

Chicago, IL  
May 2022 – Aug 2022

- Engineered backend ETL pipelines integrating data from 25+ enterprise sources into Azure SQL and Azure Data Lake.
- Developed automated workflows for ingestion, cleansing, and aggregation, supporting distributed analytics systems.

## Projects

**University Course Planning/Review Platform (Class Best):** Built a multi-service data platform enabling student schedule matching, course overlap detection, and behavioral insights. Developed ETL pipelines in Python to process thousands of course records, user preferences, and time-series activity logs. Built a responsive frontend in React + Tailwind with interactive dashboards, heatmaps, and calendar visualizations.

**Configuration Fuzzer:** Developed a configuration fuzzing tool for OSS-Fuzz, automating build generation to identify critical compile-time configurations and improve code coverage, uncovering previously untested code paths and increasing bug detection effectiveness.

**Ray Tracing Engine:** Created an interactive WebGL-based ray tracing engine in JavaScript, featuring realistic reflections, dynamic lighting, and customizable environment maps. Implemented shaders and user controls for rendering techniques, scene adjustments, and interactivity.